Abdul Moiz

AI / ML Engineer

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Portfolio | LinkedIn | GitHub

Profile

Software Engineer building end-to-end AI systems: explainable loan-risk prediction, fine-tuned coding assistance, and medical evidence retrieval. Experience spans model evaluation, inference APIs, testing, and Docker packaging.

Education

University of Karachi (UBIT) | Karachi, Pakistan
Bachelor in Software Engineering | Expected Dec 2027 | CGPA: **3.30 / 4.00**

Academic Experience

Student Developer - Image Dehazing Jan - Apr 2025
University of Karachi (UBIT) | Supervised by Dr. Humera Bashir | GitHub

- Trained a **1,761-parameter AOD-Net in PyTorch** on 35 O-HAZE scenes; selected checkpoints using 5 validation scenes.
- Measured **17.93 dB PSNR and 0.737 SSIM** on **5 held-out scenes** (384px max); hazy-input PSNR: 14.13 dB.
- Built reproducible training, inference, evaluation, and image-comparison workflows; verified **10 automated tests**.

Projects

Loan Risk Assessment System Feb - Mar 2026
Explainable credit-risk prediction | GitHub | Live Demo

- Built a Dockerized FastAPI/Streamlit loan-risk demo with ONNX inference, borrower segmentation, plain-English SHAP explanations, and account-scoped PostgreSQL records.
- Used shared profile-group splits and training-only preprocessing on 32,408 cleaned records; LightGBM achieved **0.9498 ROC-AUC and 0.8375 F1** on **4,865 held-out records**.
- Random Forest achieved **R2 0.9070** and **RMSE 0.9746 percentage points** on **4,401 held-out records**, estimating interest rates conditional on lender-assigned grade.

CodeMentor-LLM Apr - Jun 2026
Fine-tuned coding assistant | GitHub | Live Demo

- Fine-tuned **Llama 3.2 3B** with QLoRA SFT on **5,000 examples over 3 epochs** on Colab T4; DPO on **200 pairs** for 1 epoch; tracked experiments in Weights & Biases and retained fixed train/validation/test splits.
- Benchmarked base, SFT and DPO on **50 held-out examples**; DPO **ROUGE-L 0.2077** versus base 0.1807 and **BERTScore F1 0.8491** versus SFT 0.8504; reported text similarity separately from functional application tests.
- Deployed **Modal A10G in erence** behind a FastAPI/Streamlit Hugging Face Space with streaming and private history. Passed **85 automated tests**; checked live generation, saved history, logout and login; added GitHub Actions checks.

MedRAG Jul - Sep 2026
Medical evidence assistant / Local Docker application | GitHub | Live Demo

- Built a multilingual medical assistant with source citations, retrieving from **22,696 indexed chunks** across PubMed, WHO guidelines, and OpenFDA.
- Combined hybrid retrieval, reranking, and Neo4j graph enrichment; reduced pipeline latency by **49.8-66.5%** on three warmed local queries using ONNX, preserving top-five retrieval order.
- Added sign-in, saved chats, and chat-scoped PDF uploads; packaged FastAPI, Streamlit, and databases with Docker and tested authentication, isolation, and persistence.

Technical Skills

Programming & data: Python, SQL, Pandas, NumPy, scikit-learn
ML & LLMs: LightGBM, SHAP, PyTorch, Transformers, QLoRA, SFT, DPO
Retrieval & storage: Qdrant, BM25, Neo4j, PostgreSQL
Application delivery: FastAPI, Streamlit, Docker, ONNX Runtime, GitHub Actions, pytest